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## Computervision - Report

### Overview

The following outputs are generated during the preprocessing stage of a Virtual Try-On system. These outputs help the AI understand the human body, clothing regions, and pose structure before fitting new clothes onto the person.

### Pipeline Flow Diagram

Human Parsing



Generate Parse Map



Create Agnostic Mask



Generate Agnostic Image



Pose Detection



OpenPose    DensePose



Cloth Mask Generation



Final Virtual Try-On Model

## **1. Original Image**

This is the input image of the person given to the system. It contains the complete appearance of the user including clothes, body posture, and background.



### **Purpose**

It acts as the base image for generating all other outputs in the pipeline.

### **Where it is used**

Used throughout the Virtual Try-On system for pose detection, segmentation, parsing, and cloth fitting.

## 2. Agnostic-v3.2

The Agnostic image removes or hides the original clothes worn by the person while preserving body shape, face, hair, arms, and legs.



### Purpose

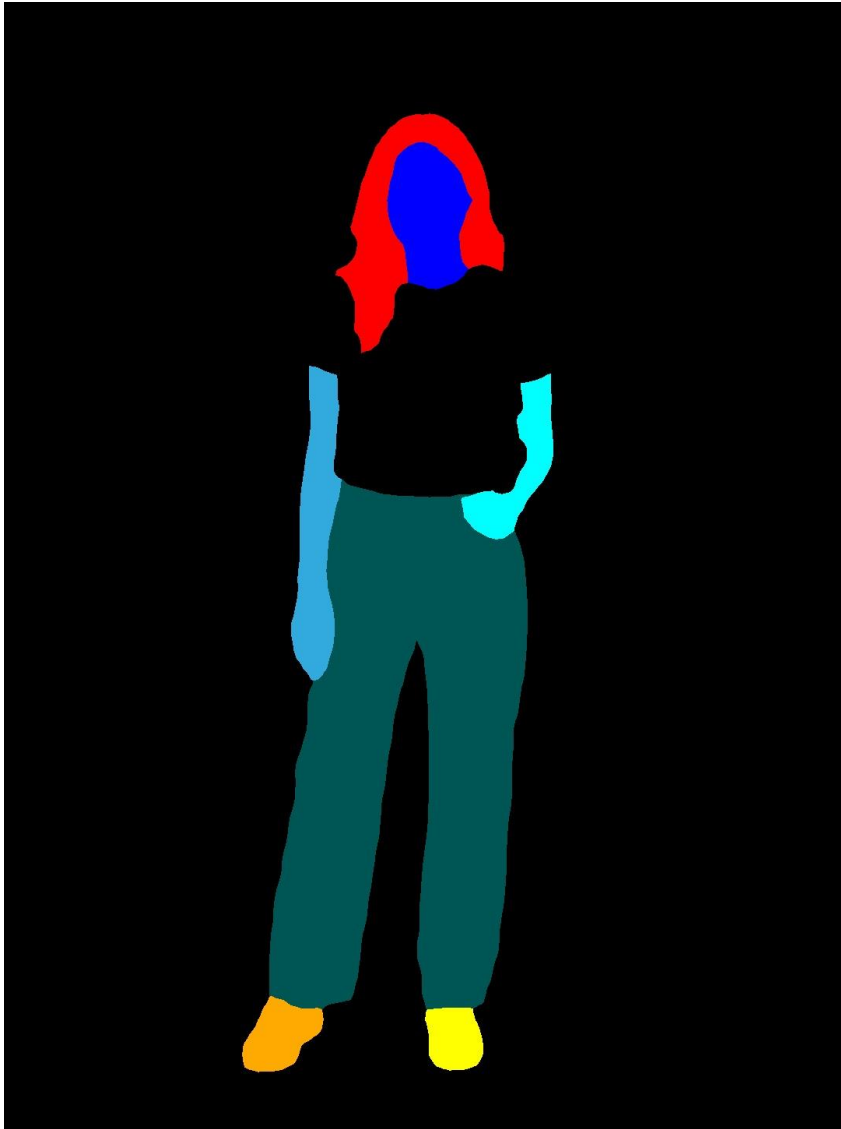
It helps the AI model focus only on the body structure so that new clothes can be applied naturally without interference from existing clothes.

### Where it is used

Used in Virtual Try-On models such as VITON-HD, IDM-VTON, and HR-VITON for realistic clothing replacement.

### 3. Parse-Agnostic-v3.2

This output removes clothing regions from the human parsing map while keeping important body parts like face, arms, and legs.



#### Purpose

It provides clean body-part segmentation information without old clothing details.

#### Where it is used

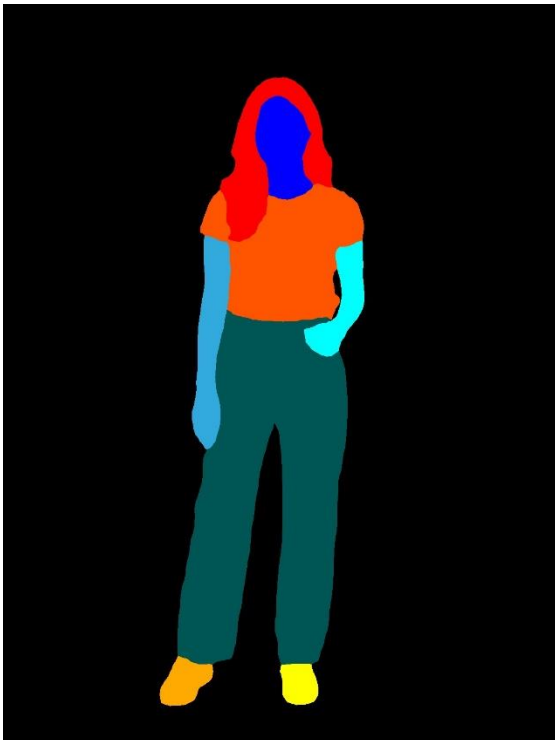
Used by the try-on model to understand body regions and accurately place new garments.

## 4. Human Parse

Human parsing divides the human body into different semantic regions such as hair, face, upper clothes, pants, arms, and legs using color labels.

For example:

- Blue → Face
- Orange → Upper clothes
- Green → Pants
- Black → Background



### Purpose

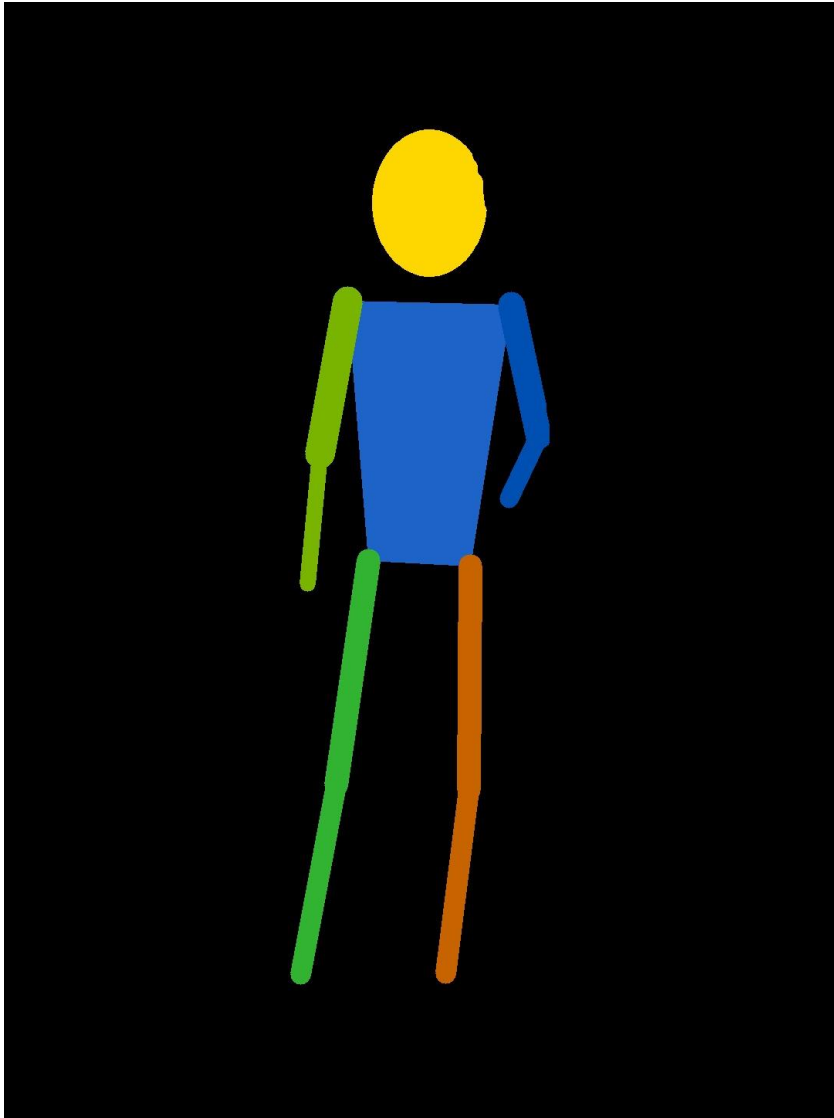
It helps the AI understand which pixels belong to each body part.

### Where it is used

Used in image segmentation, virtual try-on systems, fashion AI, and body analysis applications.

## 5. Dense Pose

Dense Pose maps different body regions and body surfaces in detail. It estimates the structure and orientation of the body.



### Purpose

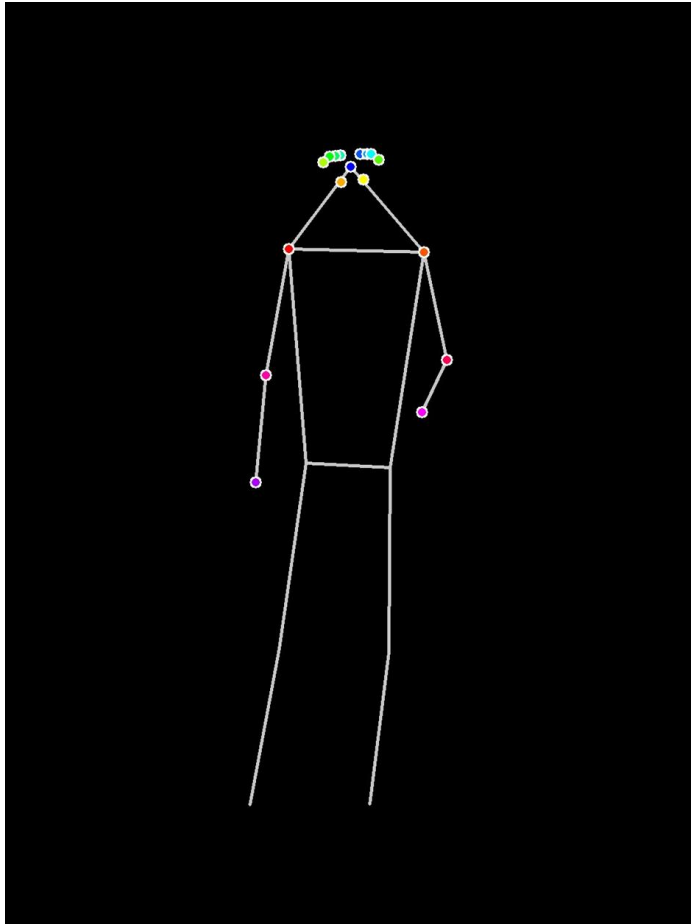
It helps in understanding body geometry and alignment for accurate cloth fitting.

### Where it is used

Used in Virtual Try-On, motion tracking, body reconstruction, animation, and augmented reality applications.

## 6. Open Pose

Open Pose detects key body joints such as shoulders, elbows, wrists, hips, knees, and ankles and connects them using skeleton lines.



### Purpose

It helps identify the body posture and orientation of the person.

### Where it is used

Used in pose estimation, fitness tracking, action recognition, gesture detection, animation, and Virtual Try-On systems.

## 7. Cloth Mask

The cloth mask generates a binary mask of the clothing item where white represents the cloth region and black represents the background.



### Purpose

It helps the AI identify the exact shape and boundary of the garment.

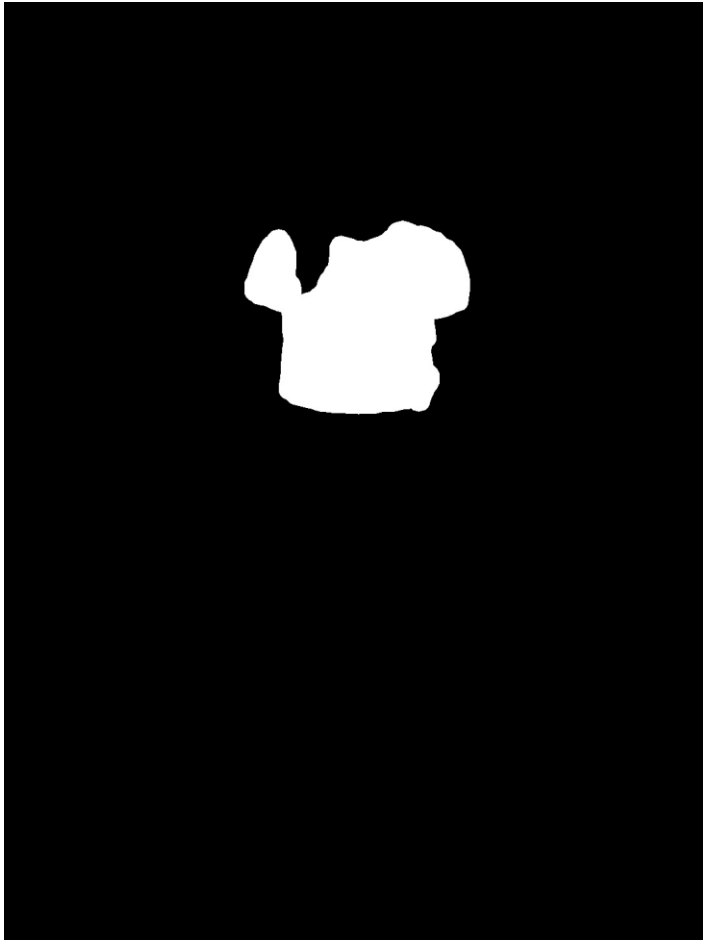
### Where it is used

Used for cloth warping, cloth alignment, segmentation, and accurate overlay in Virtual Try-On models.



## 8. Agnostic Mask

The agnostic mask highlights only the upper-body clothing region that needs to be removed or replaced during the virtual try-on process.



### Purpose

It helps the model identify the exact clothing area for garment replacement while preserving other body regions.

### Where it is used

Used in image synthesis and cloth replacement stages of Virtual Try-On systems.

### Tools Used

- SegFormer Human Parsing Model
- OpenCV Morphological Operations

## **Conclusion**

This project demonstrates the preprocessing pipeline used in AI-based Virtual Try-On systems. Different computer vision techniques such as human parsing, pose estimation, agnostic generation, cloth extraction, and masking help the system understand the human body structure and clothing regions accurately.

Human parsing identifies body parts, OpenPose and DensePose estimate body posture and orientation, while agnostic images and masks remove the original clothing information. Cloth extraction and cloth masks isolate the garment for accurate fitting.

Together, these outputs improve realistic garment transfer and enhance applications in online fashion shopping, augmented reality, and AI-powered styling systems.